

PRAKTIKUM BASIS DATA

MODUL 3

Scripting SQL Basic



Oleh:

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103122400060

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

TP MODUL 3

Aturan Tugas Pendahuluan :

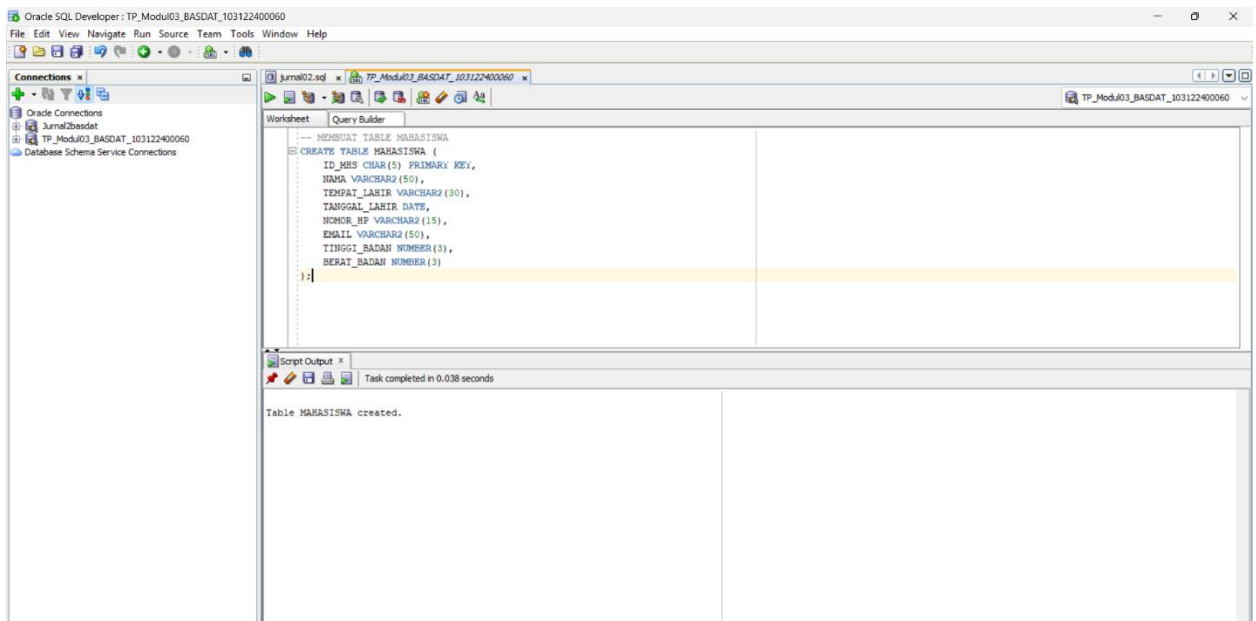
1. Buatlah table menggunakan LiveSQL ataupun tools lainnya.
2. **Screenshoot** query dan hasil perubahan pada table.
3. Dilarang **plagiarisme**.

Soal Tugas Pedahuluan :

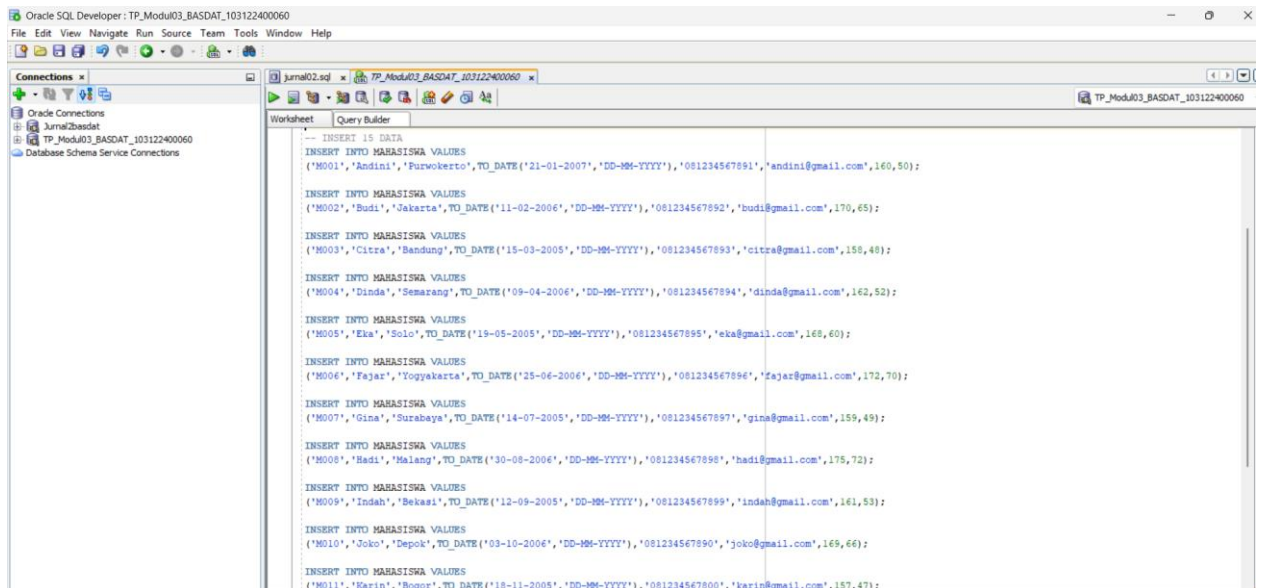
1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.
2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.
3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
 - a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)
 - b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))
 - c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))
 - d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

Jawaban:

1. Table mahasiswa



2. Insert data pada table nomor 1 sebanyak 15 rows.



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Connections x

Oracle Connections

Jurnal2basdat

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Database Schema Service Connections

Worksheet Query Builder

TP_Modul03_BASDAT_103122400060

INSERT INTO MAHASISWA VALUES
(
'M012','Lutfi','Cilacap',TO_DATE('27-12-2006','DD-MM-YYYY'),'081234567801','lutfi@gmail.com',173,71);

INSERT INTO MAHASISWA VALUES
(
'M013','Maya','Tasikmalaya',TO_DATE('05-01-2005','DD-MM-YYYY'),'081234567802','maya@gmail.com',163,55);

INSERT INTO MAHASISWA VALUES
(
'M014','Nanda','Tegal',TO_DATE('08-02-2006','DD-MM-YYYY'),'081234567803','nanda@gmail.com',171,68);

INSERT INTO MAHASISWA VALUES
(
'M015','Oki','Pemalang',TO_DATE('22-03-2005','DD-MM-YYYY'),'081234567804','oki@gmail.com',167,62);

-- MENAMPILKAN DATA
SELECT * FROM MAHASISWA;

Script Output x Query Result x

All Rows Fetched: 15 in 0.006 seconds

ID_MHS	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN	
1	M001	Andini	Purwokerto	21-JAN-07	081234567891	andini@gmail.com	160	50
2	M002	Budi	Jakarta	11-FEB-06	081234567892	budi@gmail.com	170	65
3	M003	Citra	Bandung	15-MAR-05	081234567893	citra@gmail.com	158	48
4	M004	Dinda	Semarang	09-APR-06	081234567894	dinda@gmail.com	162	52
5	M005	Eka	Solo	19-MAY-05	081234567895	eka@gmail.com	168	60
6	M006	Fajar	Yogyakarta	25-JUN-06	081234567896	fajar@gmail.com	172	70
7	M007	Gina	Surabaya	14-JUL-05	081234567897	gina@gmail.com	159	49
8	M008	Hadi	Malang	30-AUG-06	081234567898	hadi@gmail.com	175	72
9	M009	Indah	Bekasi	12-SEP-05	081234567899	indah@gmail.com	161	53
10	M010	Joko	Depok	03-OCT-06	081234567890	joko@gmail.com	169	66
11	M011	Karin	Bojort	18-NOV-05	081234567800	karin@gmail.com	157	47
12	M012	Lutfi	Cilacap	27-DEC-06	081234567801	lutfi@gmail.com	173	71
13	M013	Maya	Tasikmalaya	05-JAN-05	081234567802	maya@gmail.com	163	55
14	M014	Nanda	Tegal	08-FEB-06	081234567803	nanda@gmail.com	171	68
15	M015	Oki	Pemalang	22-MAR-05	081234567804	oki@gmail.com	167	62

(1 more...)

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
 - a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)

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Connections x

Oracle Connections

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Database Schema Service Connections

Worksheet Query Builder

```

SELECT
  NAMA,
  LOWER(NAMA) AS LOWER_NAMA,
  UPPER(NAMA) AS UPPER_NAMA,
  INITCAP(NAMA) AS INITCAP_NAMA,
  CONCAT(NAMA, ' Mahasiswa') AS CONCAT_NAMA,
  LENGTH(NAMA) AS LENGTH_NAMA,
  SUBSTR(NAMA,1,4) AS SUBSTR_NAMA,
  LPAD(NAMA,20,'') AS LPAD_NAMA,
  RPAD(NAMA,20,'') AS RPAD_NAMA,
  INSTR(EMAIL,'@') AS POSISI_AT,
  REPLACE(EMAIL,'mail','mail') AS REPLACE_EMAIL
FROM MAHASISWA;

```

Script Output x Query Result x

All Rows Fetched: 15 in 0.004 seconds

ID	NAMA	LOWER_NAMA	UPPER_NAMA	INITCAP_NAMA	CONCAT_NAMA	LENGTH_NAMA	SUBSTR_NAMA	LPAD_NAMA	RPAD_NAMA	POSISI_AT	REPLACE_EMAIL
1	Andini	andini	ANDINI	Andini	Andini Mahasiswa	6	Andi	*****Andini	Andini*****	7	andini@mail.com
2	Budi	budi	BUDI	Budi	Budi Mahasiswa	4	Budi	*****Budi	Budi*****	5	budi@mail.com
3	Citra	citra	CITRA	Citra	Citra Mahasiswa	5	Citr	*****Citra	Citra*****	6	citra@mail.com
4	Dinda	dinda	DINDA	Dinda	Dinda Mahasiswa	5	Dind	*****Dinda	Dinda*****	6	dinda@mail.com
5	Eka	eka	EKA	Eka	Eka Mahasiswa	3	Eka	*****Eka	Eka*****	4	eka@mail.com
6	Fajar	fajar	FAJAR	Fajar	Fajar Mahasiswa	5	Faja	*****Fajar	Fajar*****	6	fajar@mail.com
7	Gina	gina	GINA	Gina	Gina Mahasiswa	4	Gina	*****Gina	Gina*****	5	gina@mail.com
8	Hadi	hadi	HADI	Hadi	Hadi Mahasiswa	4	Hadi	*****Hadi	Hadi*****	5	hadi@mail.com
9	Indah	indah	INDAH	Indah	Indah Mahasiswa	5	Inda	*****Indah	Indah*****	6	indah@mail.com
10	Joko	joko	JOKO	Joko	Joko Mahasiswa	4	Joko	*****Joko	Joko*****	5	joko@mail.com
11	Karin	karin	KARIN	Karin	Karin Mahasiswa	5	Kari	*****Karin	Karin*****	6	karin@mail.com
12	Lutfi	lutfi	LUTFI	Lutfi	Lutfi Mahasiswa	5	Lutf	*****Lutfi	Lutfi*****	6	lutfi@mail.com
13	Maya	maye	MAYA	Maye	Maye Mahasiswa	4	Maya	*****Maye	Maya*****	5	maye@mail.com
14	Nanda	nanda	NANDA	Nanda	Nanda Mahasiswa	5	Nand	*****Nanda	Nanda*****	6	nanda@mail.com
15	Oki	oki	OKI	Oki	Oki Mahasiswa	3	Oki	*****Oki	Oki*****	4	oki@mail.com

(1 more...)

Line 16 Column 16 Insert Modified Windows: CRAP

- b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))

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Connections x

- Oracle Connections
- Jurnal2basdat
- TP_Modul03_BASDAT_103122400060
- Database Schema Service Connections

Worksheet Query Builder

```

SELECT
  NAMA,
  TO_CHAR(TANGGAL_LAHIR,'MM') AS BULAN ANGKA,
  TO_CHAR(TANGGAL_LAHIR,'MON') AS BULAN SINGKAT,
  TO_CHAR(TANGGAL_LAHIR,'MONTH') AS BULAN LENGKAP,
  TO_CHAR(TANGGAL_LAHIR,'DD') AS HARI,
  TO_CHAR(TANGGAL_LAHIR,'DY') AS HARI SINGKAT,
  TO_CHAR(TANGGAL_LAHIR,'YYYY') AS TAHUN,
  TO_CHAR(TANGGAL_LAHIR,'YY') AS TAHUN PENDEK,
  ADD_MONTHS(TANGGAL_LAHIR,3) AS TAMBAH_3_BULAN,
  LAST_DAY(TANGGAL_LAHIR) AS HARI TERAKHIR,
  MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR) AS SELISIH_BULAN,
  ROUND(TANGGAL_LAHIR,'YEAR') AS ROUND TAHUN
FROM MABASISWA;

```

Script Output x Query Result x

All Rows Fetched: 15 in 0.005 seconds

	NAMA	BULAN ANGKA	BULAN SINGKAT	BULAN LENGKAP	HARI	HARI SINGKAT	TAHUN	TAHUN PENDEK	TAMBAH_3_BULAN	HARI TERAKHIR	SELISIH_BULAN
1	Andini	01	JAN	JANUARY	21	SUN	2007	07	21-APR-07	31-JAN-07	231.9008225059737156511350059737156511
2	Budi	02	FEB	FEBRUARY	11	SAT	2006	06	11-MAY-06	28-FEB-06	243.22340315113500597371565113500597371
3	Citra	03	MAR	MARCH	15	TUE	2005	05	15-JUN-05	31-MAR-05	254.09437089307048984468339307048984468
4	Dinda	04	APR	APRIL	09	SUN	2006	06	09-JUL-06	30-APR-06	241.287919280167264038231780167264038231
5	Eka	05	MAY	MAY	19	THU	2005	05	19-AUG-05	31-MAY-05	251.965338635005973715651135005973715651
6	Fajar	06	JUN	JUNE	25	SUN	2006	06	25-SEP-06	30-JUN-06	238.77179024790919952210274790919952210
7	Gina	07	JUL	JULY	14	THU	2005	05	14-OCT-05	31-JUL-05	250.12662895758661887694145758661887694
8	Badri	08	AUG	AUGUST	30	WED	2006	06	30-NOV-06	31-AUG-06	236.61049992532855436081242532855436081
9	Indah	09	SEP	SEPTEMBER	12	MON	2005	05	12-DEC-05	30-SEP-05	248.1911450866188769414575866188769414
10	Joko	10	OCT	OCTOBER	03	TUE	2006	06	03-JAN-07	31-OCT-06	235.481467667264038231780167264038231
11	Karin	11	NOV	NOVEMBER	18	FRI	2005	05	18-FEB-06	30-NOV-05	245.9975966995221027479091995221027479
12	Lufti	12	DEC	DECEMBER	27	WED	2006	06	27-MAR-07	31-DEC-06	232.7072741188769414575866188769414575
13	Maya	01	JAN	JANUARY	05	WED	2005	05	05-APR-05	31-JAN-05	256.41695153823178016726403823178016726
14	Nanda	02	FEB	FEBRUARY	08	WED	2006	06	08-MAY-06	28-FEB-06	243.3201773446833930704898446833930704
15	Oki	03	MAR	MARCH	22	TUE	2005	05	22-JUN-05	31-MAR-05	253.8685644414575866188769414575866188

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Connections x

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- Database Schema Service Connections

Worksheet Query Builder

```

SELECT
  NAMA,
  TO_CHAR(TANGGAL_LAHIR,'MM') AS BULAN ANGKA,
  TO_CHAR(TANGGAL_LAHIR,'MON') AS BULAN SINGKAT,
  TO_CHAR(TANGGAL_LAHIR,'MONTH') AS BULAN LENGKAP,
  TO_CHAR(TANGGAL_LAHIR,'DD') AS HARI,
  TO_CHAR(TANGGAL_LAHIR,'DY') AS HARI SINGKAT,
  TO_CHAR(TANGGAL_LAHIR,'YYYY') AS TAHUN,
  TO_CHAR(TANGGAL_LAHIR,'YY') AS TAHUN PENDEK,
  ADD_MONTHS(TANGGAL_LAHIR,3) AS TAMBAH_3_BULAN,
  LAST_DAY(TANGGAL_LAHIR) AS HARI TERAKHIR,
  MONTHS_BETWEEN(SYSDATE,TANGGAL_LAHIR) AS SELISIH_BULAN,
  ROUND(TANGGAL_LAHIR,'YEAR') AS ROUND TAHUN
FROM MABASISWA;

```

Script Output x Query Result x

All Rows Fetched: 15 in 0.005 seconds

	NGKA	BULAN SINGKAT	BULAN LENGKAP	HARI	HARI SINGKAT	TAHUN	TAHUN PENDEK	TAMBAH_3_BULAN	HARI TERAKHIR	SELISIH_BULAN	ROUND TAHUN
1	JAN	JANUARY	21	SUN	2007	07		21-APR-07	31-JAN-07	231.900822505973715651135005973715651135	01-JAN-07
2	FEB	FEBRUARY	11	SAT	2006	06		11-MAY-06	28-FEB-06	243.223403151135005973715651135005973716	01-JAN-06
3	MAR	MARCH	15	TUE	2005	05		15-JUN-05	31-MAR-05	254.094370893070489844683393070489844683	01-JAN-05
4	APR	APRIL	09	SUN	2006	06		09-JUL-06	30-APR-06	241.287919280167264038231780167264038232	01-JAN-06
5	MAY	MAY	19	THU	2005	05		19-AUG-05	31-MAY-05	251.965338635005973715651135005973715651	01-JAN-05
6	JUN	JUNE	25	SUN	2006	06		25-SEP-06	30-JUN-06	238.771790247909199522102747909199522103	01-JAN-06
7	JUL	JULY	14	THU	2005	05		14-OCT-05	31-JUL-05	250.126628957586618876941457586618876941	01-JAN-06
8	AUG	AUGUST	30	WED	2006	06		30-NOV-06	31-AUG-06	236.610499925328554360812425328554360812	01-JAN-07
9	SEP	SEPTEMBER	12	MON	2005	05		12-DEC-05	30-SEP-05	248.191145086618876941457586618876941458	01-JAN-06
10	OCT	OCTOBER	03	TUE	2006	06		03-JAN-07	31-OCT-06	235.48146766726403823178016726403823178	01-JAN-06
11	NOV	NOVEMBER	18	FRI	2005	05		18-FEB-06	30-NOV-05	245.997596699522102747909199522102747909	01-JAN-06
12	DEC	DECEMBER	27	WED	2006	06		27-MAR-07	31-DEC-06	232.707274118876941457586618876941457587	01-JAN-07
13	JAN	JANUARY	05	WED	2005	05		05-APR-05	31-JAN-05	256.416951538231780167264038231780167264	01-JAN-05
14	FEB	FEBRUARY	08	WED	2006	06		08-MAY-06	28-FEB-06	243.32017734468339307048984468339307049	01-JAN-06
15	MAR	MARCH	22	TUE	2005	05		22-JUN-05	31-MAR-05	253.86856444145758661887694145758661887701	01-JAN-05

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

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Connections x

- Oracle Connections
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 - Database Schema Service Connections

Worksheet Query Builder

```

-- FUNGSI KONVERSI
--
SELECT
  NAMA,
  TO_NUMBER('170') + TINGGI_BADAN AS HASIL_TO_NUMBER,
  TO_CHAR(TANGGAL_LAHIR, 'DD MONTH YYYY') AS TO_CHAR_TANGGAL,
  TO_CHAR(BERAT_BADAN, '999') AS TO_CHAR_BERAT,
  TO_DATE('21-01-2007', 'DD-MM-YYYY') AS TO_DATE_CONTOH
FROM MAHASISWA;

```

Script Output x Query Result x All Rows Fetched: 15 in 0.003 seconds

	NAMA	HASIL_TO_NUMBER	TO_CHAR_TANGGAL	TO_CHAR_BERAT	TO_DATE_CONTOH
1	Andini	330	21 JANUARY 2007	50	21-JAN-07
2	Budi	340	11 FEBRUARY 2006	65	21-JAN-07
3	Citra	328	15 MARCH 2005	48	21-JAN-07
4	Dinda	332	05 APRIL 2006	52	21-JAN-07
5	Eka	338	19 MAY 2005	60	21-JAN-07
6	Fajar	342	25 JUNE 2006	70	21-JAN-07
7	Gina	329	14 JULY 2005	49	21-JAN-07
8	Hadi	345	30 AUGUST 2006	72	21-JAN-07
9	Indah	331	12 SEPTEMBER 2005	53	21-JAN-07
10	Joko	339	03 OCTOBER 2006	66	21-JAN-07
11	Karin	327	18 NOVEMBER 2005	47	21-JAN-07
12	Lutfi	343	27 DECEMBER 2006	71	21-JAN-07
13	Maya	333	05 JANUARY 2005	55	21-JAN-07
14	Nanda	341	08 FEBRUARY 2006	68	21-JAN-07
15	Oki	337	22 MARCH 2005	62	21-JAN-07

(1 more...) Line 107 Column 16 Insert Modified Windows: CRAP

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

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Connections x

- Oracle Connections
 - JurnalBasdat
 - TP_Modul03_BASDAT_103122400060
 - Database Schema Service Connections

Worksheet Query Builder

```

-- FUNGSI NUMERIK
--
SELECT
  NAMA,
  ABS(BERAT_BADAN - 60) AS ABS_CONTOH,
  MOD(TINGGI_BADAN, 2) AS MOD_CONTOH,
  FLOOR(BERAT_BADAN / 3) AS FLOOR_CONTOH,
  POWER(2, 3) AS POWER_CONTOH,
  ROUND(BERAT_BADAN / 3.2) AS ROUND_CONTOH,
  CEIL(TINGGI_BADAN / 7) AS CEIL_CONTOH
FROM MAHASISWA;

```

Script Output x Query Result x All Rows Fetched: 15 in 0.003 seconds

	NAMA	ABS_CONTOH	MOD_CONTOH	FLOOR_CONTOH	POWER_CONTOH	ROUND_CONTOH	CEIL_CONTOH
1	Andini	10	0	16	8	16.67	23
2	Budi	5	0	21	8	21.67	25
3	Citra	12	0	16	8	16	23
4	Dinda	8	0	17	8	17.33	24
5	Eka	0	0	20	8	20	24
6	Fajar	10	0	23	8	23.33	25
7	Gina	11	1	16	8	16.33	23
8	Hadi	12	1	24	8	24	25
9	Indah	7	1	17	8	17.67	23
10	Joko	6	1	22	8	22	25
11	Karin	13	1	15	8	15.67	23
12	Lutfi	11	1	23	8	23.67	25
13	Maya	5	1	18	8	18.33	24
14	Nanda	8	1	22	8	22.67	25
15	Oki	2	1	20	8	20.67	24

(1 more...) Line 121 Column 16 Insert Modified Windows: CRAP

